ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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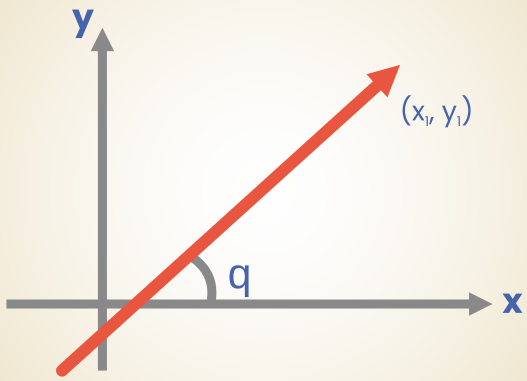
MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

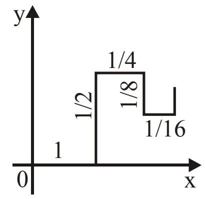
Straight Line

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. A particle begins at the origin and moves successively in the following manner as shown,



1 unit to the right, 1/2 unit up, 1/4 unit to the right,
1/8 unit down, 1/16 unit to the right etc.

The length of each move is half the length of the previous move and movement continues in the 'zigzag' manner infinitely. The co-ordinates of the point to which the 'zigzag' converges is -

(A) (4/3, 2/3) (B) (4/3,2/5) (C) (3/2,2/3) (D) (2,2/5)

SL0001

2. Suppose ABC is a triangle with 3 acute angles A,B and C. The point whose coordinates are (cosB – sinA, sinB – cosA) can be in the -

(A) first and 2nd quadrant (B) second the 3rd quadrant
(C) third and 4th quadrant (D) second quadrant only

SL0002

3. The length of a line segment AB is 10 units. If the coordinates of one extremity are (2,–3) and the abscissa of the other extremity is 10 then the sum of all possible values of the ordinate of the other extremity is -

(A) 3 (B) –4 (C) 12 (D) –6

SL0006

4. Consider the points P(2,–4); Q(4,–2) and R(7,1). The points P,Q,R -

(A) form an equilateral triangle
(B) form a right angled triangle
(C) form an isosceles triangle which is not equilateral
(D) are collinear.

SL0007

5. If in triangle ABC, A ≡ (1,10), circumcenter ≡ $\left(-\frac{1}{3},\frac{2}{3}\right)$ and orthocenter ≡ $\left(\frac{11}{3},\frac{4}{3}\right)$ then the co-ordinates of mid-point of side opposite to A is-

(A) (1,–11/3) (B) (1,5) (C) (1,–3) (D) (1,6)

SL0014

6. The co-ordinates of the orthocentre of the triangle bounded by the lines, $4x - 7y + 10 = 0$; $x + y = 5$ and $7x + 4y = 15$ is-

(A) (2,1) (B) (–1,2) (C) (1,2) (D) (1,–2)

SL0015

7. A triangle has two of its vertices at (0,1) and (2,2) in the cartesian plane. Its third vertex lies on the x-axis. If the area of the triangle is 2 square units then the sum of the possible abscissae of the third vertex, is-

(A) –4 (B) 0 (C) 5 (D) 6

SL0016

8. A line passes through (2,2) and cuts a triangle of area 9 square units from the first quadrant. The sum of all possible values for the slope of such a line, is-

(A) –2.5 (B) –2 (C) –1.5 (D) –1

SL0017

9. Let A(2,–3) and B(–2,1) be vertices of a ΔABC. If the centroid of ΔABC moves on the line $2x + 3y = 1$, then the locus of the vertex C is-

(A) $2x + 3y = 9$ (B) $2x - 3y = 7$

(C) $3x + 2y = 5$ (D) $3x - 2y = 3$

SL0018

10. A stick of length 10 units rests against the floor and a wall of a room. If the stick begins to slide on the floor then the locus of its middle point is :

(A) $x^2 + y^2 = 2.5$ (B) $x^2 + y^2 = 25$

(C) $x^2 + y^2 = 100$ (D) none

SL0019

11. A and B are any two points on the positive x and y axis respectively satisfying 2(OA) + 3(OB) = 10. If P is the middle point of AB then the locus of P is-

(A) $2x + 3y = 5$ (B) $2x + 3y = 10$

(C) $3x + 2y = 5$ (D) $3x + 2y = 10$

SL0020

12. Given the points A(0,4) and B(0,–4), the equation of the locus of the point P such that $|AP - BP| = 6$ is -

(A) $9x^2 - 7y^2 + 63 = 0$ (B) $9x^2 - 7y^2 - 63 = 0$

(C) $7x^2 - 9y^2 + 63 = 0$ (D) $7x^2 - 9y^2 - 63 = 0$

SL0022

13. If x_1,y_1 are the roots of $x^2 + 8x - 20 = 0$, x_2,y_2 are the roots of $4x^2 + 32x - 57 = 0$ and x_3,y_3 are the roots of $9x^2 + 72x - 112 = 0$, then the points, (x_1,y_1), (x_2,y_2) and (x_3,y_3) -

(A) are collinear

(B) form an equilateral triangle

(C) form a right angled isosceles triangle

(D) are concyclic

SL0027

14. The area of the parallelogram formed by the lines 3x + 4y = 7a; 3x + 4y = 7b; 4x + 3y = 7c and 4x + 3y = 7d is-

(A) $\frac{|(a-b)(c-d)|}{7}$

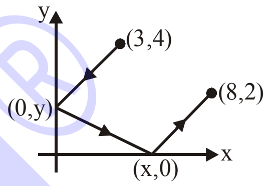
(B) $|(a-b)(c-d)|$

(C) $\frac{|(a-b)(c-d)|}{49}$

(D) $7|(a-b)(c-d)|$

SL0029

15. Suppose that a ray of light leaves the point (3,4), reflects off the y-axis towards the x-axis, reflects off the x-axis, and finally arrives at the point (8,2). The value of x, is -



(A) $x = 4\frac{1}{2}$

(B) $x = 4\frac{1}{3}$

(C) $x = 4\frac{2}{3}$

(D) $x = 5\frac{1}{3}$

SL0032

16. m,n are integer with 0 < n < m. A is the point (m,n) on the cartesian plane. B is the reflection of A in the line y = x. C is the reflection of B in the y-axis, D is the reflection of C in the x-axis and E is the reflection of D in the y-axis. The area of the pentagon ABCDE is -

(A) 2m(m + n)　(B) m(m + 3n)　(C) m(2m + 3n)　(D) 2m(m + 3n)

SL0033

17. If the lines $\left.\begin{array}{l}\lambda x+(\sin\alpha)y+\cos\alpha=0\\ x+(\cos\alpha)y+\sin\alpha=0\\ x-(\sin\alpha)y+\cos\alpha=0\end{array}\right]$ pass through the same point where $\alpha \in R$ then λ lies in the interval -

(A) [–1,1]　(B) $\left[-\sqrt{2},\sqrt{2}\right]$　(C) [–2,2]　(D) $(-\infty,\infty)$

SL0034

18. If the straight lines, ax + amy + 1 = 0, bx + (m + 1) by + 1 = 0 and cx + (m + 2)cy + 1 = 0, m ≠ 0 are concurrent then a,b,c are in :

(A) A.P. only for m = 1

(B) A.P. for all m

(C) G.P. for all m

(D) H.P. for all m

SL0035

19. Family of lines represented by the equation (cosθ + sinθ)x + (cosθ – sinθ)y – 3(3 cosθ + sinθ) = 0 passes through a fixed point M for all real values of θ. The reflection of M in the line x – y = 0, is-

(A) (6,3)　(B) (3,6)　(C) (–6,3)　(D) (3,–6)

SL0036

20. Consider a parallelogram whose sides are represented by the lines $2x + 3y = 0$; $2x + 3y - 5 = 0$; $3x - 4y = 0$ and $3x - 4y = 3$. The equation of the diagonal not passing through the origin, is-

(A) $21x - 11y + 15 = 0$ (B) $9x - 11y + 15 = 0$

(C) $21x - 29y - 15 = 0$ (D) $21x - 11y - 15 = 0$

SL0037

21. If the equation $ax^2 - 6xy + y^2 + 2gx + 2fy + c = 0$ represents a pair of lines whose slopes are m and m^2, then sum of all possible values of a is-

(A) 17 (B) –19 (C) 19 (D) –17

SL0040

22. Let $S = \{(x,y)|\ x^2 + 2xy + y^2 - 3x - 3y + 2 = 0\}$, then S -

(A) consists of two coincident lines.

(B) consists of two parallel lines which are not coincident.

(C) consists of two intersecting lines.

(D) is a parabola.

SL0041

23. The line x= c cuts the triangle with corners (0,0); (1,1) and (9,1) into two region. For the area of the two regions to be the same c must be equal to-

(A) 5/2 (B) 3 (C) 7/2 (D) 3 or 15

SL0062

24. If m and b are real numbers and $mb > 0$, then the line whose equation is $y = mx + b$ cannot contain the point-

(A) (0,2009) (B) (2009,0)

(C) (0,–2009) (D) (20,–100)

SL0063

25. In a triangle ABC, side AB has the equation $2x + 3y = 29$ and the side AC has the equation, $x + 2y = 16$. If the mid- point of BC is (5,6) then the equation of BC is :

(A) $x - y = -1$ (B) $5x - 2y = 13$

(C) $x + y = 11$ (D) $3x - 4y = -9$

SL0064

26. The vertex of the right angle of a right angled triangle lies on the straight line $2x - y - 10 = 0$ and the two other vertices, at points (2,–3) and (4,1) then the area of triangle in sq. units is-

(A) $\sqrt{10}$ (B) 3 (C) $\frac{33}{5}$ (D) 11

SL0065

27. A triangle ABC is formed by the lines $2x - 3y - 6 = 0$; $3x - y + 3 = 0$ and $3x + 4y - 12 = 0$. If the points $P(\alpha,0)$ and $Q(0,\beta)$ always lie on or inside the ΔABC, then :

(A) $\alpha \in [-1,2]$ and $\beta \in [-2,3]$
(B) $\alpha \in [-1,3]$ and $\beta \in [-2,4]$
(C) $\alpha \in [-2,4]$ and $\beta \in [-3,4]$
(D) $\alpha \in [-1,3]$ and $\beta \in [-2,3]$

SL0066

28. The co-ordinates of a point P on the line $2x - y + 5 = 0$ such that $|PA - PB|$ is maximum where A is $(4,-2)$ and B is $(2,-4)$ will be :

(A) $(11,27)$
(B) $(-11,-17)$
(C) $(-11,17)$
(D) $(0,5)$

SL0067

29. The line $(k + 1)^2x + ky - 2k^2 - 2 = 0$ passes through a point regardless of the value k. Which of the following is the line with slope 2 passing through the point ?

(A) $y = 2x - 8$
(B) $y = 2x - 5$
(C) $y = 2x - 4$
(D) $y = 2x + 8$

SL0068

30. If the straight lines joining the origin and the points of intersection of the curve

$5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$

are equally inclined to the x- axis then the value of k :

(A) is equal to 1
(B) is equal to -1
(C) is equal to 2
(D) does not exist in the set of real numbers

SL0069

EXERCISE (O-2)

Multiple Correct Answer Type

1. The x-coordinates of the vertices of a square of unit area are the roots of the equation $x^2-3|x|+2=0$ and the y-coordinates of the vertices are the roots of the equation $y^2-3y+2=0$ then the possible vertices of the square is/are-

(A) (1,1), (2,1), (2,2), (1,2)　(B) (–1,1), (–2,1), (–2,2), (–1,2)

(C) (2,1), (1,–1),(1,2),(2,2)　(D) (–2,1), (–1,–1),(–1,2), (–2,2)

SL0071

2. Three vertices of a triangle are A(4,3); B(1,–1) and C(7,k). Value(s) of k for which centroid, orthocentre, incentre and circumcentre of the ΔABC lie on the same straight line is/are-

(A) 7　(B) –1　(C) –19/8　(D) none

SL0072

3. Line $\frac{x}{a}+\frac{y}{b}=1$ cuts the co-ordinate axes at A(a,0) and B(0,b) and the line $\frac{x}{a'}+\frac{y}{b'}=-1$ at A'(–a',0) and B'(0,–b'). If the points A,B,A',B' are concyclic then the orthocentre of the triangle ABA' is -

(A) (0,0)　(B) (0,b')　(C) $\left(0,\frac{aa'}{b}\right)$　(D) $\left(0,\frac{bb'}{a}\right)$

SL0073

4. If the vertices P,Q,R of a triangle PQR are rational points, which of the following points of the triangle PQR is/are always rational point(s) ?

(A) centroid　(B) incentre　(C) circumcentre　(D) orthocentre

SL0074

5. The area of triangle ABC is 20 square units. The co-ordinates of vertex A are (–5,0) and B are (3,0). The vertex C lies on the line, x – y = 2. The co-ordinates of C are -

(A) (5,3)　(B) (–3,–5)　(C) (–5,–7)　(D) (7,5)

SL0075

6. Consider the equation $y-y_1=m(x-x_1)$. If m and x_1 are fixed and different lines are drawn for different values of y_1, then :

(A) the lines will pass through a fixed point

(B) there will be a set of parallel lines

(C) all the lines intersect the line $x=x_1$

(D) all the lines will be parallel to the line $y=x_1$.

SL0076

7. If one vertex of an equilateral triangle of side 'a' lies at the origin and the other lies on the line $x-\sqrt{3}y=0$ then the co-ordinates of the third vertex are :

(A) (0,a) (B) $\left(\frac{\sqrt{3}a}{2},-\frac{a}{2}\right)$ (C) (0,–a) (D) $\left(-\frac{\sqrt{3}a}{2},\frac{a}{2}\right)$

SL0077

8. Let B(1,–3) and D(0,4) represent two vertices of rhombus ABCD in (x,y) plane, then coordinates of vertex A if $\angle BAD = 60°$ can be equal to-

(A) $\left(\frac{1-7\sqrt{3}}{2},\frac{1-\sqrt{3}}{2}\right)$ (B) $\left(\frac{1+7\sqrt{3}}{2},\frac{1+\sqrt{3}}{2}\right)$

(C) $\left(\frac{1-14\sqrt{3}}{2},\frac{1-2\sqrt{3}}{2}\right)$ (D) $\left(\frac{1+14\sqrt{3}}{2},\frac{1+2\sqrt{3}}{2}\right)$

SL0078

9. The sides of a triangle are the straight lines x + y = 1; 7y = x and $\sqrt{3}y+x=0$. Then which of the following is an interior point of the triangle ?

(A) circumcentre (B) centroid (C) incentre (D) orthocentre

SL0079

10. A line passes through the origin and makes an angle of $\pi/4$ with the line x – y + 1 = 0. Then :

(A) equation of the line is x = 0

(B) the equation of the line is y = 0

(C) the point of intersection of the line with the given line is (–1,0)

(D) the point of intersection of the line with the given line is (0,1)

SL0080

11. Equation of a straight line passing through the point (2,3) and inclined at an angle of arc tan $\frac{1}{2}$ with the line y + 2x = 5 is-

(A) y = 3 (B) x = 2

(C) 3x + 4y – 18 = 0 (D) 4x + 3y – 17 = 0

SL0081

12. If $a^2 + 9b^2 - 4c^2 = 6ab$ then the family of lines ax + by + c = 0 are concurrent at :

(A) (1/2, 3/2) (B) (–1/2, –3/2) (C) (–1/2, 3/2) (D) (1/2, –3/2)

SL0082

13. The lines L_1 and L_2 denoted by $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ intersect at the point P and have gradients m_1 and m_2 respectively. The acute angles between them is θ. Which of the following relations hold good ?

(A) $m_1 + m_2 = 5/4$

(B) $m_1m_2 = 3/8$

(C) acute angle between L_1 and L_2 is $\sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$.

(D) sum of the abscissa and ordinate of the point P is –1.

SL0083

14. Let $y = m_1x$ & $y = m_2x$ be the equations of the lines joining the origin to the points of trisection of the portion of the line $3x + y = 12$ intercepted between the axes, then -

(A) $m_1 + m_2 = \frac{21}{2}$ (B) $|m_1 - m_2| = \frac{9}{2}$

(C) $m_1.m_2 = 9$ (D) $\frac{m_1}{m_2} = 6$

SL0167

15. If the equation $ax^2 - 6xy + y^2 + bx + cy + d = 0$ represents pair of lines whose slopes are m and m^2, then value of a is/are -

(A) $a = -8$ (B) $a = 8$ (C) $a = 27$ (D) $a = -27$

SL0168

16. The triangle formed by the lines $(5x + 12y)^2 = (12x - 5y)^2$ and $5x + 12y = 13$ will be -

(A) acute angled (B) right angled

(C) obtuse angled (D) isosceles

SL0169

17. If the slope of one of the line represent by $\lambda x^2 - 2xy + y^2 = 0$ is square of the other, then λ can be -

(A) –8 (B) 8 (C) –1 (D) 1

SL0170

18. The point $(\alpha^2, \alpha - 1)$ lie on same side of $x - 6y + 2 = 0$ as that of (–1, 1), then α may be -

(A) $\frac{5}{2}$ (B) 3 (C) $\frac{9}{2}$ (D) $\frac{7}{3}$

SL0171

19. The position of the point (1, 1) w.r.t. the lines $L_1 \equiv 2x + 3y + 7 = 0$ and $L_2 \equiv 4x + 6y - 19 = 0$ is such that :

(A) it is below L_2 (B) it is above L_1

(C) it is between L_1 & L_2 (D) it is above both L_1 & L_2

SL0172

20. $2x^2 - 3xy + y^2 + 5x - 4y + 3 = 0$ represent two straight lines of slope m_1 and m_2. Then choose correct options -

(A) harmonic mean of m_1 and m_2 is $\frac{4}{3}$

(B) arithmetic mean of m_1 and m_2 is 3

(C) point of intersection of these lines is $(0, -1)$

(D) point of intersection of these lines is $(-2, -1)$

SL0173

21. Two adjacent sides of a rhombus are $2x + 3y = a - 5$ and $3x + 2y = 4 - 2a$ and its diagonals intersect at the point (1,2), then a can be -

(A) -16 (B) 16 (C) $-\frac{10}{3}$ (D) $\frac{10}{3}$

SL0174

22. Point P(2,3) lies on the line $4x + 3y = 17$. Then find the co-ordinates of points farthest from the line which are at 5 units distance from the P.

(A) (6, 6) (B) (6, –6) (C) (2, 0) (D) (–2, 0)

SL0175

23. If one side of a square is parallel to $3x - 4y = 0$ & its area being 16 while centre being (1, 1), then find equation of sides of square.

(A) $3x - 4y + 11 = 0$ (B) $3x - 4y - 9 = 0$

(C) $4x + 3y + 3 = 0$ (D) $4x + 3y - 17 = 0$

SL0176

24. Let L be the line $2x + y = 2$. If the axes are rotated by 45°, then the intercepts made by the line L on the new axes are respectively

(A) $\sqrt{2}$ and 1 (B) 1 and $\sqrt{2}$

(C) $2\sqrt{2}$ and $2\sqrt{2}/3$ (D) $2\sqrt{2}/3$ and $2\sqrt{2}$

SL0177

25. Find the equations of the sides of a triangle having (4, –1) as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.

(A) $2x - y + 3 = 0$ (B) $x + 2y - 6 = 0$

(C) $2x + y - 7 = 0$ (D) $x - 2y - 6 = 0$

SL0178

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question Nos. 1 to 3

Consider two points $A \equiv (1,2)$ and $B \equiv (3,-1)$. Let M be a point on the straight line $L \equiv x + y = 0$.

1. If M be a point on the line L = 0 such that AM + BM is minimum, then the reflection of M in the line x = y is -

(A) (1,–1) (B) (–1,1) (C) (2,–2) (D) (–2,2)

SL0086

2. If M be a point on the line L = 0 such that |AM – BM| is maximum, then the distance of M from $N \equiv (1,1)$ is-

(A) $5\sqrt{2}$ (B) 7 (C) $3\sqrt{5}$ (D) 10

SL0086

3. If M be a point on the line L = 0 such that |AM – BM| is minimum, then the area of ΔAMB equals-

(A) $\frac{13}{4}$ (B) $\frac{13}{2}$ (C) $\frac{13}{6}$ (D) $\frac{13}{8}$

SL0086

Paragraph for Question Nos. 4 to 5

Let ABCD is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that AE= AF. Let P be a point inside the square ABCD.

4. The maximum possible area of quadrilateral CDFE is -

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{8}$

SL0084

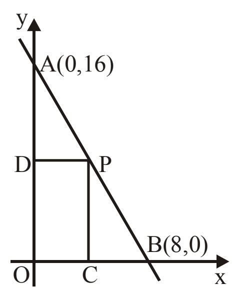
5. The value of $(PA)^2 - (PB)^2 + (PC)^2 - (PD)^2$ can not be equal to -

(A) 3 (B) 2 (C) 1 (D) 0

SL0084

Paragraph for Question Nos. 6 to 8

In the diagram, a line is drawn through the points A(0,16) and B(8,0). Point P is chosen in the first quadrant on the line through A and B. Points C and D are chosen on the x and y axis respectively, so that PDOC is a rectangle.



6. If perpendicular distance of the line AB from the point (2, 2) is $\lambda\sqrt{5}$ then value of λ is-

SL0085

7. Sum of the coordinates of the point P if PDOC is a square is -

SL0085

8. Number of possible ordered pair(s) of all positions of the point P on AB so that the area of the rectangle PDOC is 30 sq. units, is-

SL0085

Matrix Match Type

9. **List-I** **List-II**

	List-I		List-II
(I)	If the incentre of the triangle formed by the coordinate axes and the line $3x+4y=12$ is at (a, b), then a + b equals	(P)	1
(II)	The line $2x-3y=6$ is shifted by 1 unit closer to the origin, the equation of line in new position is $2x-3y=6+k\sqrt{13}$, then value of $\|k\|$ is	(Q)	2
(III)	Sum of the abscissa of points on the line $x-y=3$ which lie at a unit distance from the line $4x-3y=12$ is	(R)	3
(IV)	The integral part of 'a' for which the point (2, a) lies between the parallel lines $x+2y=4$ and $x+2y=5$ is	(S)	6

Which of the following is correct ?

(A) I $\to$ Q; II $\to$ R; III $\to$ P; IV $\to$ S (B) I $\to$ P; II $\to$ Q; III $\to$ R; IV $\to$ P

(C) I $\to$ Q; II $\to$ P; III $\to$ S; IV $\to$ P (D) I $\to$ S; II $\to$ R; III $\to$ P; IV $\to$ Q

SL0179

10. In ΔABC, let slope of side AB is $\frac{1}{3}$ & internal angle bisector AD is $3x=4y$. Also mirror image of vertex B w.r.t. line AD is (13, 16) & point D is (12, 9) (where D lies on BC).

	Column-I		Column-II
(A)	Co-ordinates of vertex 'B' are (a, b), then (a – b) is equal to	(P)	$\frac{13}{9}$
(B)	Slope of side AC, is	(Q)	$\frac{25}{4}$
(C)	x-intercept of external angle bisector of vertex 'A' is	(R)	11
(D)	Abscissa of point C is $\frac{a}{b}$ where a & b are coprime numbers, then $\|a-b\|$ is equal to	(S)	13
		(T)	15

SL0180

EXERCISE (O-4)

Numerical Grid Type

1. Let O(0, 0), A(6, 0) and B(3, $\sqrt{3}$) be the vertices of ΔOAB. Let R be the region consisting of all those points P inside ΔOAB which satisfy d(P, OA) $\le$ minimum (d(P,OB), d(P,AB)) where d(P,OA), d(P,OB) and d(P,AB) represent the distance of P from the sides OA,OB and AB respectively. If the area of region R is $9(a-\sqrt{b})$ where a and b are coprime, then find the value of (a + b)

SL0094

2. A triangle has side lengths 18, 24 and 30. Find the area of the triangle whose vertices are the incentre, circumcentre and centroid of the triangle. (In unit)

SL0095

3. The line 3x + 2y = 24 meets the y-axis at A & the x-axis at B. The perpendicular bisector of AB meets the line through (0, –1) parallel to x-axis at C. Find the area of the triangle ABC. (In Sq. unit)

SL0098

4. Consider a ΔABC whose sides AB, BC and CA are represented by the straight lines 2x + y = 0, x + py = q and x – y = 3 respectively. The point P is (2, 3).

(a) If P is the centroid, then find the value of (p + q).

(b) If P is the orthocentre, then find the value of (p + q).

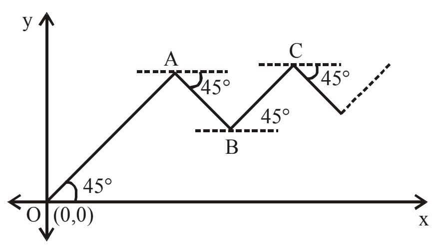
(c) If P is the circumcentre, then find the value of (p + q).

SL0117

5. The two line pairs $y^2-4y+3=0$ and $x^2+4xy+4y^2-5x-10y+4=0$ enclose a 4 sided convex polygon find **(i)** area of the polygon; **(ii)** length of its diagonals.

SL0123

6. Point O, A, B, C .... are shown in figure where OA = 2AB = 4BC = ... so on. Let A be the centroid of a triangle whose orthocentre and circumcenter are (2, 4) and $\left(\frac{7}{2}, \frac{5}{2}\right)$ respectively. If an insect starts moving from the point O(0,0) along the straight line in zig-zag fashion and terminates ultimately at point P(α, β) then find the value of ($\alpha + \beta$)



SL0112

7. If the straight line drawn through the point $P(\sqrt{3}, 2)$ & inclined at an angle $\frac{\pi}{6}$ with the x-axis, meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q. Find the length PQ. (in unit)

SL0099

8. The sides of a triangle have the combined equation $x^2 - 3y^2 - 2xy + 8y - 4 = 0$. The third side, which is variable always passes through the point (–5, –1). If the range of values of the slope of the third line so that the origin is an interior point of the triangle, lies in the interval (a, b), then find $\left(a + \frac{1}{b^2}\right)$.

SL0122

9. The interior angle bisector of angle A for the triangle ABC whose coordinates of the vertices are A(–8, 5); B(–15, –19) and C(1, – 7) has the equation ax + 2y + c = 0. Find a + c.

SL0114

10. If algebraic sum of lengths of perpendiculars drawn from vertices of a hexagon ABCDEF, A(–2,–1), B(0,–2), C(4,0), D(3,2), E(2,3), F(–1,4) on a variable line ax + by + c = 0 is zero, then variable line always passes through a fixed point (p,q), then the value of p + q is

SL0181

EXERCISE (JM)

1. The x-coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as (0, 1)(1, 1) and (1, 0) is : **[JEE-MAIN 2013]**

(1) $2+\sqrt{2}$ (2) $2-\sqrt{2}$ (3) $1+\sqrt{2}$ (4) $1-\sqrt{2}$

SL0127

2. A light ray emerging from the point source placed at P(1, 3) is reflected at a point Q in the axis of *x*. If the reflected ray passes through the point R(6, 7), then the abscissa of Q is :

[JEE-MAIN Online 2013]

(1) 3 (2) $\frac{7}{2}$ (3) 1 (4) $\frac{5}{2}$

SL0128

3. If the three lines x–3y = p, ax + 2y = q and ax + y = r from a right – angled triangle then:

[JEE-MAIN Online 2013]

(1) $a^2 - 6a - 12 = 0$ (2) $a^2 - 9a + 12 = 0$

(3) $a^2 - 9a + 18 = 0$ (4) $a^2 - 6a - 18 = 0$

SL0129

4. If the extremities of the base of an isosceles triangle are the points (2a, 0) and (0, a) and the equation of one of the sides isx = 2a, then the area of the triangle, in square units, is :

[JEE-MAIN Online 2013]

(1) $\frac{5}{2}a^2$ (2) $\frac{5}{4}a^2$ (3) $\frac{25a^2}{4}$ (4) $5a^2$

SL0131

5. Let θ_1 be the angle between two lines $2x + 3y + c_1 = 0$ and $-x + 5y + c_2 = 0$, andθ_2 be the angle between two lines $2x + 3y + c_1 = 0$ and $-x + 5y + c_3 = 0$, where c_1, c_2, c_3 are any real numbers :

[JEE-MAIN Online 2013]

Statement–1 : If c_2 and c_3 are proportional, then $\theta_1 = \theta_2$.

Statement–2 : $\theta_1 = \theta_2$ for all c_2 and c_3.

(1) Statement-1 is true and Statement - 2 is true, Statement-2 is not a correct explanation for Statement-1.

(2) Statement-1 is false and Statement-2 is true.

(3) Statement-1 is true and Statement-2 is false.

(4) Statement-1 is true and Statement - 2 is true, Statement-2 is a correct explanation for Statement-1.

SL0132

6. Let a, b, c and d be non-zero numbers. If the point of intersection of the lines 4ax + 2ay + c = 0 and 5bx + 2by + d = 0 lies in the fourth quadrant and is equidistant from the two axes then :

[JEE(Main)-2014]

(1) 2bc – 3ad = 0 (2) 2bc + 3ad = 0 (3) 3bc – 2ad = 0 (4) 3bc + 2ad = 0

SL0135

7. Let PS be the median of the triangle with vertices P (2, 2), Q (6, –1) and R (7, 3). The equation of the line passing through (1, –1) and parallel to PS is : **[JEE(Main)-2014]**

(1) $4x - 7y - 11 = 0$ (2) $2x + 9y + 7 = 0$

(3) $4x + 7y + 3 = 0$ (4) $2x - 9y - 11 = 0$

SL0136

8. Locus of the image of the point (2, 3) in the line $(2x - 3y + 4) + k(x - 2y + 3) = 0$, $k \in R$, is a

(1) circle of radius $\sqrt{2}$ (2) circle of radius $\sqrt{3}$

(3) straight line parallel to x-axis (4) straight line parallel to y-axis

[JEE(Main)-2015]

SL0137

9. Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at (–1, –2), then which one of the following is a vertex of this rhombus ? **[JEE(Main)-2016]**

(1) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$ (2) (–3, –9) (3) (–3, –8) (4) $\left(\frac{1}{3}, -\frac{8}{3}\right)$

SL0138

10. Let k be an integer such that triangle with vertices (k, –3k), (5, k) and (–k, 2) has area 28 sq. units. Then the orthocentre of this triangle is at the point : **[JEE(Main)-2017]**

(1) $\left(2, \frac{1}{2}\right)$ (2) $\left(2, -\frac{1}{2}\right)$ (3) $\left(1, \frac{3}{4}\right)$ (4) $\left(1, -\frac{3}{4}\right)$

SL0139

11. Let the orthocentre and centroid of a triangle be A(–3, 5) and B(3, 3) respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is:

[JEE(Main)-2018]

(1) $2\sqrt{10}$ (2) $3\sqrt{\frac{5}{2}}$ (3) $\frac{3\sqrt{5}}{2}$ (4) $\sqrt{10}$

SL0140

12. A straight line through a fixed point (2, 3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is : **[JEE(Main)-2018]**

(1) $2x + 3y = xy$ (2) $3x + 2y = xy$ (3) $3x + 2y = 6xy$ (4) $3x + 2y = 6$

SL0141

13. Let S be the set of all triangles in the xy-plane, each having one vertex at the origin and the other two vertices lie on coordinate axes with integral coordinates. If each triangle in S has area 50sq. units, then the number of elements in the set S is : **[JEE(Main)-2019]**

(1) 9 (2) 18 (3) 32 (4) 36

SL0142

14. If the line $3x + 4y - 24 = 0$ intersects the x-axis at the point A and the y-axis at the point B, then the incentre of the triangle OAB, where O is the origin, is **[JEE(Main)-2019]**

(1) (3, 4) (2) (2, 2) (3) (4, 4) (4) (4, 3)

SL0143

15. Two vertices of a triangle are (0,2) and (4,3). If its orthocentre is at the origin, then its third vertex lies in which quadrant ? **[JEE(Main)-2019]**

(1) Fourth (2) Second (3) Third (4) First

SL0144

16. Two sides of a parallelogram are along the lines, $x + y = 3$ and $x - y + 3 = 0$. If its diagonals intersect at (2,4), then one of its vertex is : **[JEE(Main)-2019]**

(1) (2,6) (2) (2,1) (3) (3,5) (4) (3,6)

SL0145

17. Slope of a line passing through P(2, 3) and intersecting the line, $x + y = 7$ at a distance of 4 units from P, is **[JEE(Main)-2019]**

(1) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$ (2) $\frac{1-\sqrt{5}}{1+\sqrt{5}}$ (3) $\frac{1-\sqrt{7}}{1+\sqrt{7}}$ (4) $\frac{\sqrt{7}-1}{\sqrt{7}+1}$

SL0147

18. A rectangle is inscribed in a circle with a diameter lying along the line $3y = x + 7$. If the two adjacent vertices of the rectangle are (–8, 5) and (6, 5), then the area of the rectangle (in sq. units) is :-

[JEE(Main)-2019]

(1) 72 (2) 84 (3) 98 (4) 56

SL0148

19. A straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x + y = 0$. Then an equation of the line L is : **[JEE(Main)-2019]**

(1) $\left(\sqrt{3}+1\right)x+\left(\sqrt{3}-1\right)y=8\sqrt{2}$ (2) $\left(\sqrt{3}-1\right)x+\left(\sqrt{3}+1\right)y=8\sqrt{2}$

(3) $\sqrt{3}x+y=8$ (4) $x+\sqrt{3}y=8$

SL0149

20. The locus of the mid-points of the perpendiculars drawn from points on the line, $x = 2y$ to the line $x = y$ is : **[JEE(Main)-2020]**

(1) $2x - 3y = 0$ (2) $7x - 5y = 0$ (3) $5x - 7y = 0$ (4) $3x - 2y = 0$

SL0150

21. Let A(1, 0), B(6, 2) and $C\left(\frac{3}{2},6\right)$ be the vertices of a triangle ABC. If P is a point inside the triangle ABC such that the triangles APC, APB and BPC have equal areas, then the length of the line segment PQ, where Q is the point $\left(-\frac{7}{6},-\frac{1}{3}\right)$, is ________. **[JEE(Main)-2020]**

SL0151

22. Let two points be A(1,–1) and B(0,2). If a point P(x',y') be such that the area of $\Delta PAB = 5$ sq. units and it lies on the line, $3x + y - 4\lambda = 0$, then a value of λ is **[JEE(Main)-2020]**

(1) 1 (2) 4 (3) 3 (4) –3

SL0152

23. Let C be the centroid of the triangle with vertices (3, –1), (1, 3) and (2, 4). Let P be the point of intersection of the lines $x + 3y - 1 = 0$ and $3x - y + 1 = 0$. Then the line passing through the points C and P also passes through the point : **[JEE(Main)-2020]**

(1) (7, 6) (2) (–9, –6) (3) (–9, –7) (4) (9, 7)

SL0153

24. Let the equation of the pair of lines, $y = px$ and $y = qx$, can be written as $(y - px)(y - qx) = 0$. Then the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$ is : **[JEE(Main)-2021]**

(1) $x^2 - 3xy + y^2 = 0$ (2) $x^2 + 4xy - y^2 = 0$

(3) $x^2 + 3xy - y^2 = 0$ (4) $x^2 - 3xy - y^2 = 0$

SL0182

25. If p and q are the lengths of the perpendiculars from the origin on the lines, $x\operatorname{cosec}\alpha - y\sec\alpha = k\cot 2\alpha$ and $x\sin\alpha + y\cos\alpha = k\sin 2\alpha$ respectively, then k^2 is equal to : **[JEE(Main)-2021]**

(1) $4p^2 + q^2$ (2) $2p^2 + q^2$ (3) $p^2 + 2q^2$ (4) $p^2 + 4q^2$

SL0183

26. Let the area of the triangle with vertices A(1, α), B(α, 0) and C(0, α) be 4 sq. units. If the point (α, –α), (–α, α) and (α^2, β) are collinear, then β is equal to **[JEE(Main)-2022]**

(A) 64 (B) –8 (C) –64 (D) 512

SL0184

27. The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 15a$ and $x - y = 3$ respectively. If its orthocentre is (2, a), $-\frac{1}{2} < a < 2$, then p is equal to

[JEE(Main)-2022]

SL0185

EXERCISE (JA)

1. Consider the lines given by

$L_1 = x + 3y - 5 = 0$

$L_2 = 3x - ky - 1 = 0$

$L_3 = 5x + 2y - 12 = 0$

Match the statements / Expression in **Column-I** with the statements / Expressions in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in OMR.

	Column-I		Column-II
(A)	L_1, L_2, L_3 are concurrent, if	(P)	$k = -9$
(B)	One of L_1, L_2, L_3 is parallel to at least one of the other two, if	(Q)	$k = -\frac{6}{5}$
(C)	L_1, L_2, L_3 form a triangle, if	(R)	$k = \frac{5}{6}$
(D)	L_1, L_2, L_3 do not form a triangle, if	(S)	$k = 5$

[JEE 2008, 6]

SL0154

2. Let P, Q, R and S be the points on the plane with position vectors $-2\hat{i} - \hat{j}$, $4\hat{i}, 3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral PQRS must be a

(A) parallelogram, which is neither a rhombus nor a rectangle

(B) square

(C) rectangle, but not a square

(D) rhombus, but not a square **[JEE 2010, 3]**

SL0155

3. A straight line L through the point (3, –2) is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersect the x-axis, then the equation of L is **[JEE 2011, 3 (–1)]**

(A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

(B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

(C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

(D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

SL0156

4. For $a > b > c > 0$, the distance between (1, 1) and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then **[JEE-Advanced 2013, 2]**

(A) $a + b - c > 0$

(B) $a - b + c < 0$

(C) $a - b + c > 0$

(D) $a + b - c < 0$

SL0157

5. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \le d_1(P) + d_2(P) \le 4$, is

[JEE(Advanced)-2014, 3]

SL0158

Question Stem for Question Nos. 6 and 7

Question Stem

Consider the lines L_1 and L_2 defined by

$L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$

For a fixed constant λ, let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2. The line $y = 2x + 1$ meets C at two points R and S, where the distance between R and S is $\sqrt{270}$.

Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'.

[JEE(Advanced)-2021]

6. The value of λ^2 is __________. **SL0165**

7. The value of D is __________. **SL0166**

ANSWER KEY

Do yourself - 1

1. $PQ = \sqrt{34}$ units **2.** $x = 6$ or $x = 0$ **3.** 11, –7

4. A **5.** A **6.** C

7. **(a)** (2,1) ; **(b)** (7,16) **8.** **(a)** 2 : 3 (internally) ; **(b)** 9 : 4 (externally) ; **(c)** 8 : 7 (internally)

9. C **10.** 1 : 2 ; Q(–5, –3) **11.** D

Do yourself - 2

1. **(a)** $\left(\frac{8}{3}, 4\right)$; **(b)** $\left(-\frac{7}{2}, \frac{17}{2}\right)$, $\frac{5\sqrt{10}+2}{2}$, **(c)** (15, – 5) **2.** B

3. C **4.** D **5.** B **6.** A **7.** B

8. **(a)** $\left(2, \frac{8}{3}\right)$; **(b)** 4 **9.** C

Do yourself - 3

1. 25 square units **2.** 132 square units **4.** A **5.** B

6. $K = 7$ or $\frac{31}{9}$ **7.** $\left(\frac{7}{2}, \frac{13}{2}\right)$ **or** $\left(-\frac{3}{2}, \frac{3}{2}\right)$

Do yourself - 4

1. $x = \pm 2$ **2.** $y = \pm x$ **3.** B **4.** B **5.** B **6.** C

7. $y = 3x$ **8.** $x^2 + y^2 = 3$ **9.** $8x^2 + 8y^2 + 6x - 36y + 27 = 0$ **10.** $x^2 = 3y^2$

Do yourself - 5

1. D **2.** D **3.** D **4.** D **5.** A **6.** C **7.** D

8. D **9.** C **10.** C **11.** D **12.** B **13.** A **14.** D

15. B **16.** D **17.** A **18.** $83x - 35y + 92 = 0$

19. $c = -4$; B(2, 0); D(4, 4) **20.** **(a)** 5; **(b)** 2; **(c)** 3/2

21. $x - 3y - 31 = 0$ or $3x + y + 7 = 0$ **22.** $x - y = 0$

Do yourself - 6

1. **(a)** $y=\frac{2x}{3}+\frac{5}{3},\frac{2}{3},\frac{5}{3}$; **(b)** $\frac{x}{(-5/2)}+\frac{y}{(5/3)}=1,-\frac{5}{2},\frac{5}{3}$;

   (c) $-\frac{2x}{\sqrt{13}}+\frac{3y}{\sqrt{13}}=\frac{5}{\sqrt{13}},\frac{5}{\sqrt{13}}$, $\alpha=\pi-\tan^{-1}\left(\frac{3}{2}\right)$

2. $13\sqrt{2}/3$ units 3. D 4. D 5. C 6. A 7. D

8. A 9. D 10. D 11. D

Do yourself - 7

1. $\theta = 135°$ or $45°$ 2. $3x + 4y = 18$ 3. $2x - 3y + 12 = 0$, $(-6, 0)$

4. **(a)** Coincident, **(b)** Parallel, **(c)** Intersecting

5. C 6. B 7. C 8. A 9. A

10. $x+5y+5\sqrt{2}=0$ or $x+5y-5\sqrt{2}=0$

Do yourself - 8

1. **(a)** 2 **(b)** 33/10 2. $\left(\frac{a}{b}\left(b\pm\sqrt{a^2+b^2}\right),0\right)$ 4. 30 5. C

6. B 7. B 8. A 9. C

Do yourself - 9

1. opposite sides of the line 3. $-y + x = 11$ 4. $\lambda = -7$

5. D 6. B 7. C 8. B

Do yourself - 10

1. (A) $\rightarrow$ (p), (B) $\rightarrow$ (s), (C) $\rightarrow$ (q), (D) $\rightarrow$ (t), (E) $\rightarrow$ (r) 2. $(7, -4)$

3. $x^2 - 4y^2 = a^2$ 4. C 5. D 6. C 7. B 8. C

9. A 10. $14x + 23y = 40$ 11. 15 12. B 13. D 14. C

Do yourself - 11

1. $x - 2y + 1 = 0$ & $2x + y - 3 = 0$; **(a)** $x - 2y + 1 = 0$; **(b)** $2x + y - 3 = 0$

2. $x + y = 2$, $x = y$ 3. $69x + 46y - 25 = 0$ 4. D 5. C 6. D

7. B 8. C 9. A 10. $x^2 + y^2 - 3x + 4y - 3 = 0$ 11. $x - 5 = 0$

Do yourself - 12

1. $x-y=0$ & $x-4y=0$ **2.** $k=-1$, or $-\frac{127}{6}$

3. $2x-5y-2=0$ & $3x+2y+3=0$; $\pm\tan^{-1}\left(\frac{19}{4}\right)$ **4.** $(-1,2), 90°$

5. B **6.** A **7.** A **8.** C **9.** A **10.** A **11.** A

12. A **13.** C **14.** **(a)** $\frac{50}{7}$; **(b)** $\frac{63}{10}$; **(c)** $\frac{3}{10}\left(8\sqrt{5}-5\sqrt{10}\right)$

Do yourself - 13

1. $\theta=\pm\tan^{-1}\frac{4}{7}$; **2.** $x^2-y^2-5xy=0$ **3.** $x+y=1;\ x+9y=1$ **6.** 2

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	D	D	A	C	A	A	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A	D	B	B	B	D	B	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	B	B	C	B	D	B	A	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	B,C	B,C	A,C,D	B,D	B,C	A,B,C,D	A,B	B,C	A,B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,C	C,D	B,C,D	B,C	B,D	B,D	A,D	A,B,D	A,B,C	A,D
Que.	21	22	23	24	25					
Ans.	A,D	A,D	A,B,C,D	C,D	A,C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	B	D	A	C	A,B,C	2.00	10.66 or 10.67	2.00	C
Q.10	A	B	C	D					
	R	P	Q	T					

EXERCISE # O-4

1. 5 **2.** 3 **3.** 91 **4.** **(a)** 74; **(b)** 50; **(c)** 47

5. **(i)** area = 6 sq. units, **(ii)** diagonals are $\sqrt{5}$ & $\sqrt{53}$ **6.** 8.00 **7.** 6.00

8. 24.00 **9.** 89.00 **10.** 2.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	3	1	4	3	2	1	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	4	2	2	4	3	2	1	3
Que.	21	22	23	24	25	26	27			
Ans.	5	3	2	3	1	C	3			

EXERCISE # JEE-ADVANCED

Que.	Q.1	A	B	C	D	
Ans.		S	P,Q	R	P,Q,S	
Que.	2	3	4	5	6	7
Ans.	A	B	A or C or A,C	6	9.00	77.14